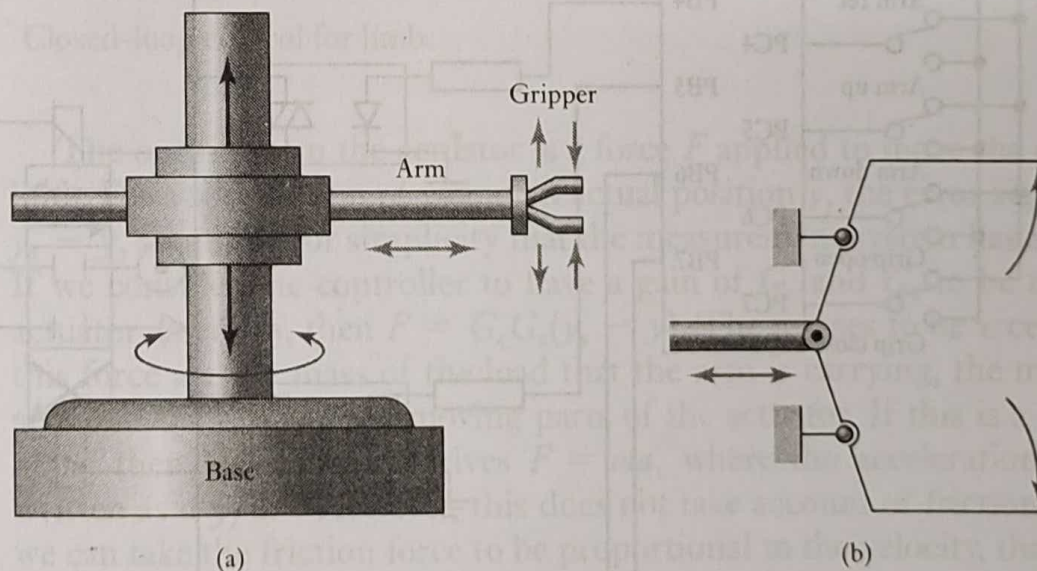


24.2.1 A pick-and-place robot

Figure 24.12(a) shows the basic form of a pick-and-place robot unit. The robot has three axes about which motion can occur: rotation in a clockwise or anti-clockwise direction of the unit on its base, arm extension or contraction and arm up or down; also the gripper can open or close. These movements can be actuated by the use of pneumatic cylinders operated by solenoid-controlled valves with limit switches to indicate when a motion is completed. Thus the clockwise rotation of the unit might result from the piston in a cylinder being extended and the anti-clockwise direction by its retraction. Likewise the upward movement of the arm might result from the piston in a linear cylinder being extended and the downward motion from it retracting; the extension of the arm by the piston in another cylinder extending and its return movement by the piston retracting. The gripper can be opened or closed by the piston in a linear cylinder extending or retracting, Figure 24.12(b) showing a basic mechanism that could be used.

Pick-and-place,



A typical program for such a robot might be:

- 1 Close an upright gripper on a component hanging from an overhead feeder.
- 2 Contract the arm so that the component is withdrawn from the feeder.
- 3 Rotate the arm in a horizontal plane so that it points in the direction of the workpiece.
- 4 Extend the arm so that the gripper is over the workpiece.
- 5 Rotate the wrist so that the component hangs downwards from the gripper.
- 6 Release the gripper so that the component falls into the required position.
- 7 Rotate the gripper into an upright position.
- 8 Contract the arm.
- 9 Rotate the arm to point towards the feeder.

Repeat the sequence for the next component.

Figure 24.13 shows how a microcontroller could be used to control solenoid valves and hence the movements of the robot unit.

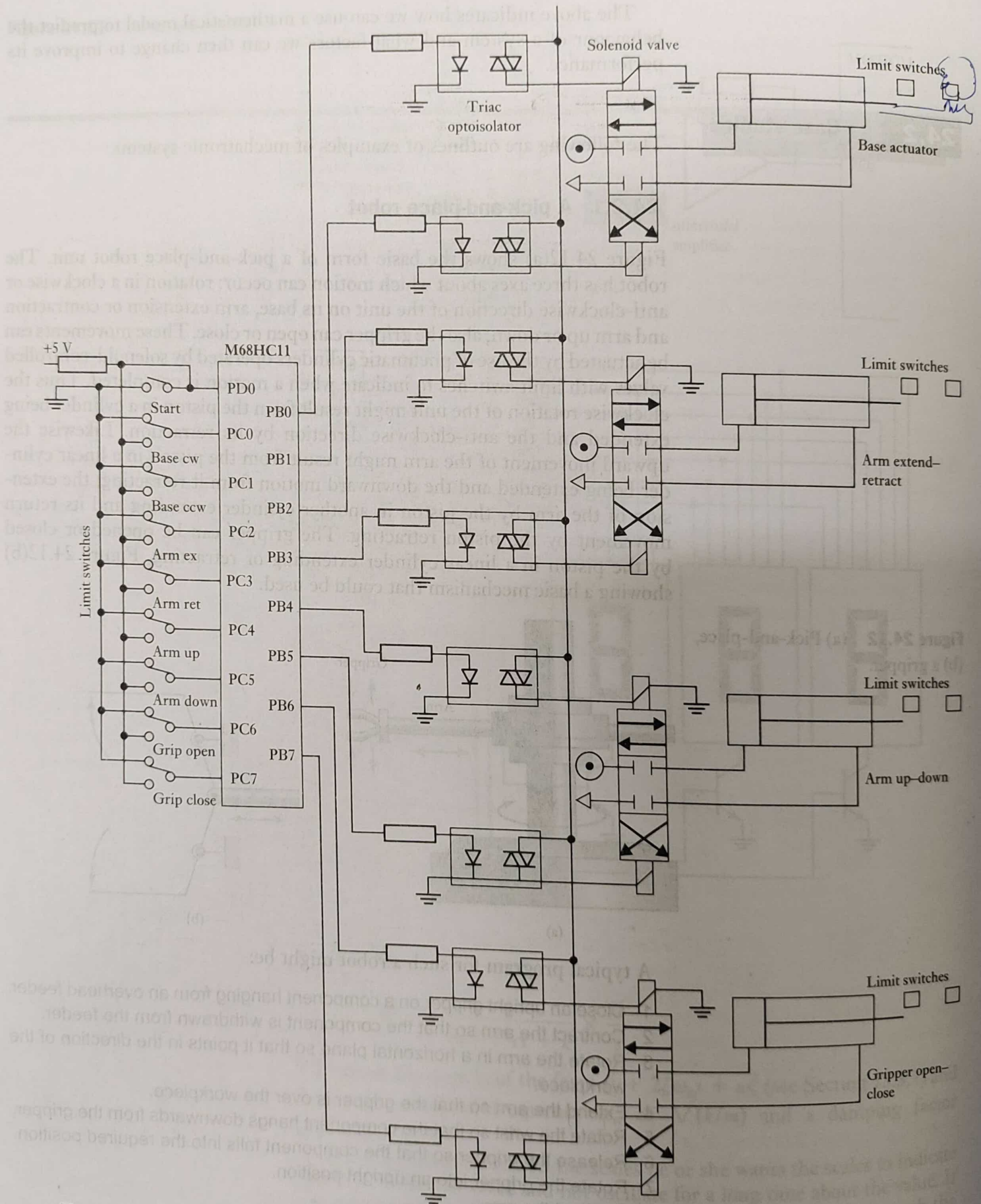


Figure 24.13 Robot control.

Hydraulic and pneumatic rams are widely used to drive robot arms since they can easily be controlled to move limbs at a relatively slow speed, while electric motors would need to operate through a gearbox.

The positions of the arm and gripper in Figure 24.13 are determined by limit switches. This means that only two positions can be accurately attained with each actuator and the positions cannot be readily changed without physically moving the positions of the switches. The arrangement is an open-loop control system. In some applications this may not be a problem.

However, it is more common to use closed-loop control with the positions of an arm and gripper being monitored by sensors and fed back to be compared in the controller with the positions required. When there is a difference from the required positions, the controller operates actuators to reduce the error. The angular position of a joint is often monitored by using an encoder (see Section 2.3.7), this being capable of high precision. Figure 24.14 shows a closed-loop arrangement that might be used for linear motion of a robot arm.

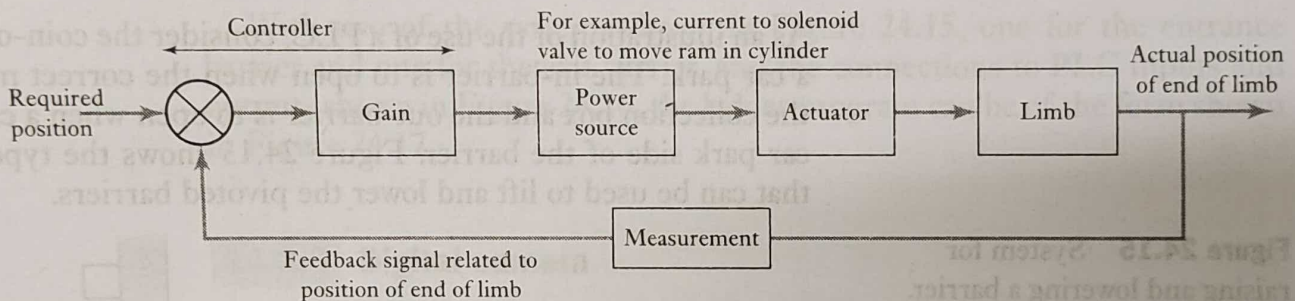


Figure 24.14 Closed-loop control for limb.

The output from the actuator is a force F applied to move the end of the limb. For a set position of y_s and an actual position y , the error signal will be $y_s - y$, assuming for simplicity that the measurement system has a gain of 1. If we consider the controller to have a gain of G_c and G_a to be that of the actuator assembly, then $F = G_c G_a (y_s - y)$. The masses to be accelerated by this force are the mass of the load that the arm is carrying, the mass of the arm and the mass of the moving parts of the actuator. If this is a total mass of m , then Newton's law gives $F = ma$, where the acceleration a can be written as d^2y/dt^2 . However, this does not take account of friction and since we can take the friction force to be proportional to the velocity, the frictional force is $k dy/dt$. Thus we can write

$$F = G_c G_a (y_s - y) = m \frac{d^2y}{dt^2} + k \frac{dy}{dt}$$

and so

$$y_s = \frac{m}{G_c G_a} \frac{d^2y}{dt^2} + \frac{k}{G_c G_a} \frac{dy}{dt} + y$$

This is a second-order differential equation and so the deflection y will be as described in Section 13.3.1 and the form it will take will depend on the damping factor. An under-damped system will have a natural angular frequency ω_n given by

$$\omega_n = \sqrt{\frac{G_c G_a}{m}}$$

This angular frequency will determine how fast the system responds to a change (see Section 12.5): the larger the angular frequency, the faster the system responds (the rise time is inversely proportional to the angular frequency). This means that increasing the controller gain or decreasing the mass can increase the speed of response. The damping factor ζ is given from the differential equation as

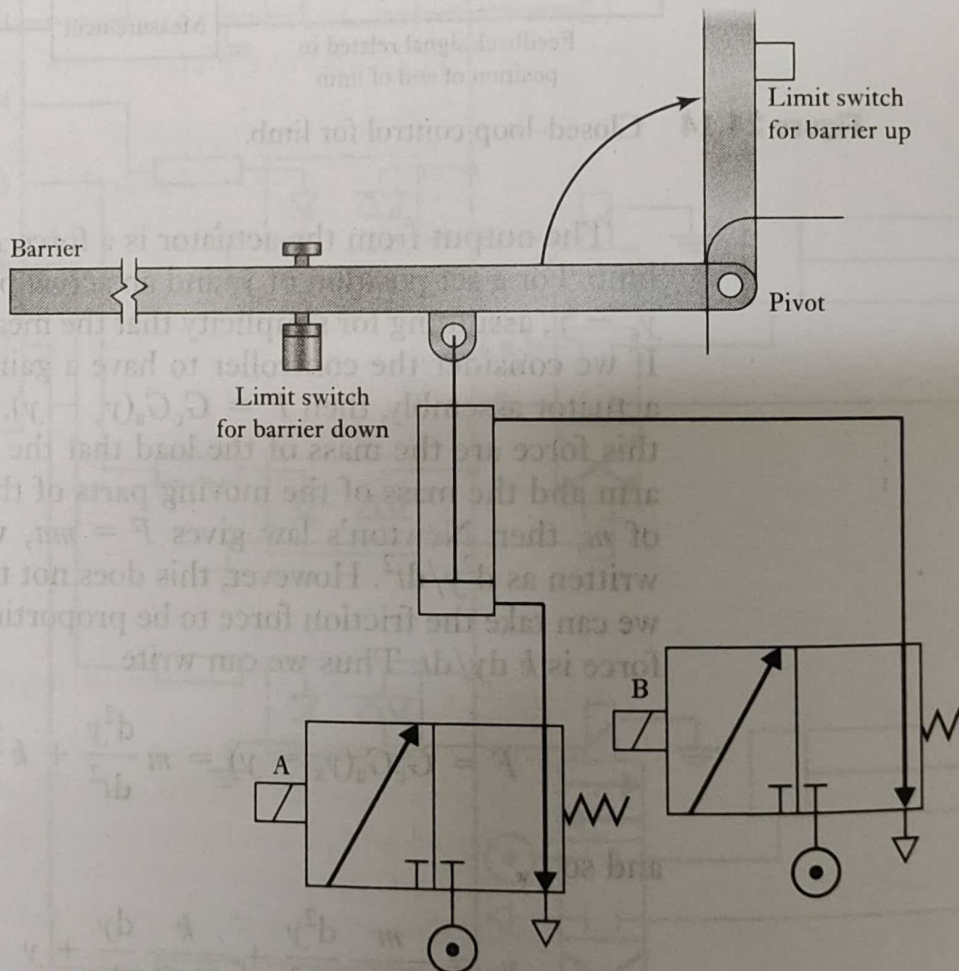
$$\zeta = \frac{k}{2\sqrt{G_c G_a m}}$$

The time taken for the oscillations to die away, i.e. the settling time (see Section 12.5), is inversely proportional to the damping factor and so, for example, increasing any part of the mass will result in a decrease in the damping factor and so the oscillations take longer to die away.

24.2.2 Car park barriers

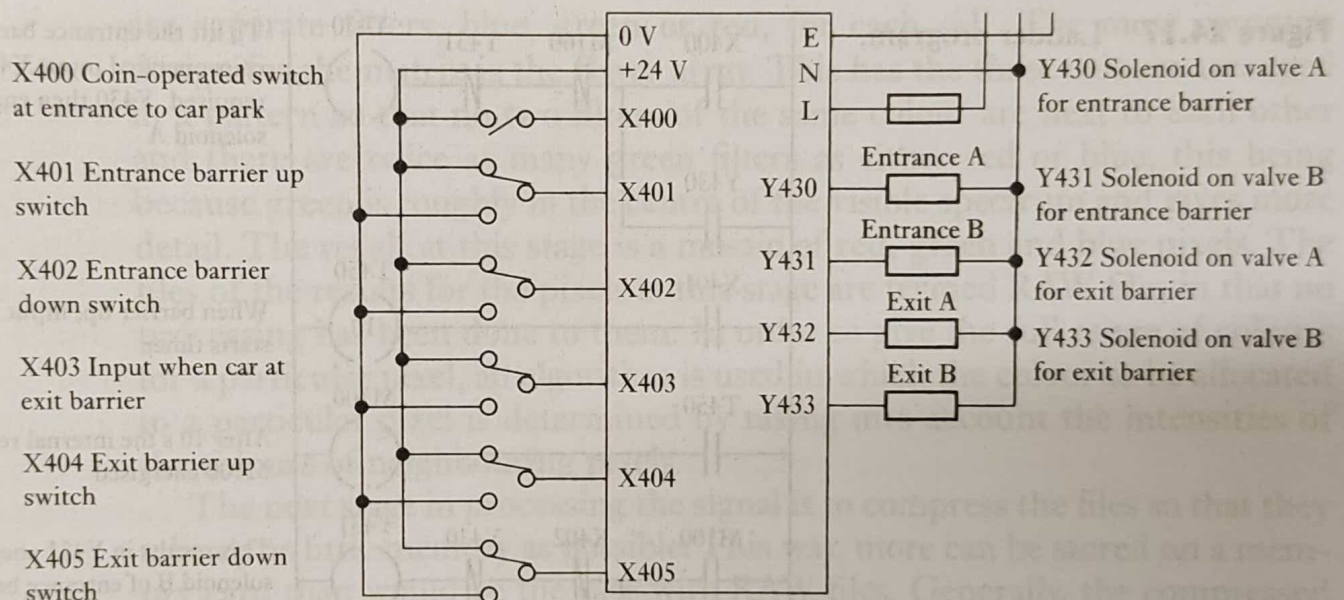
As an illustration of the use of a PLC, consider the coin-operated barriers for a car park. The in-barrier is to open when the correct money is inserted in the collection box and the out-barrier is to open when a car is detected at the car park side of the barrier. Figure 24.15 shows the types of valve systems that can be used to lift and lower the pivoted barriers.

system for
ing a barrier.



When a current flows through the solenoid of valve A, the piston in a cylinder moves upwards and causes the barrier to rotate about its pivot and raise to let a car through. When the current through the solenoid of valve A ceases, the return spring of the valve results in the valve position changing back to its original position. When the current is switched through the solenoid of valve B, the pressure is applied to lower the barrier. Limit switches are used to detect when the barrier is down and also when fully up.

PLC



With two of the systems shown in Figure 24.15, one for the entrance barrier and one for the exit barrier, and the connections to PLC inputs and outputs shown in Figure 24.16, the ladder program can be of the form shown in Figure 24.17.

This means that the output will reach the required position after just one sample. This is a much faster response. By using a high sampling rate a very fast response can be achieved. This form of response is termed a **deadbeat response**.

24.2.4 Car engine management

The modern car is likely to include many electronic control systems involving microcontrollers, the engine control system being one, its aim being to ensure that the engine is operated at its optimum settings. Figure 24.22 shows a generalised block diagram of such a system. The system consists of sensors supplying, after suitable signal conditioning, the input signals to the microcontroller and its providing output signals via drivers to actuate actuators. Figure 24.23 shows some of these elements in relation to an engine, only one cylinder being shown.

The engine speed sensor is an inductive sensor and consists of a coil for which the inductance changes as the teeth of the sensor wheel pass it and so gives an oscillating voltage. The temperature sensor is usually a thermistor. The mass air flow sensor may be a hot wire sensor; as air passes over a heated wire it will be cooled, the amount of cooling depending on the mass rate of

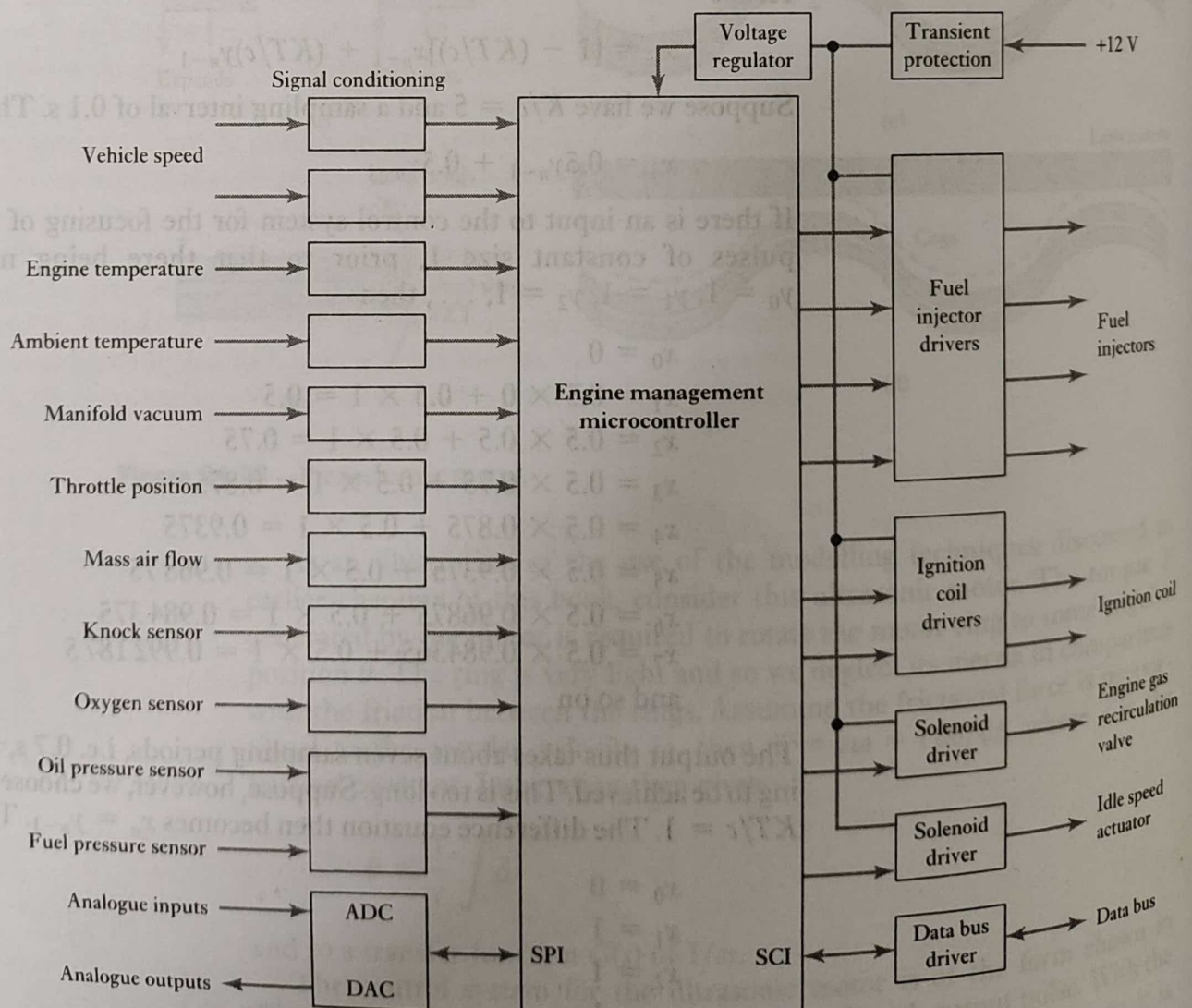


Figure 24.22 Engine management system.

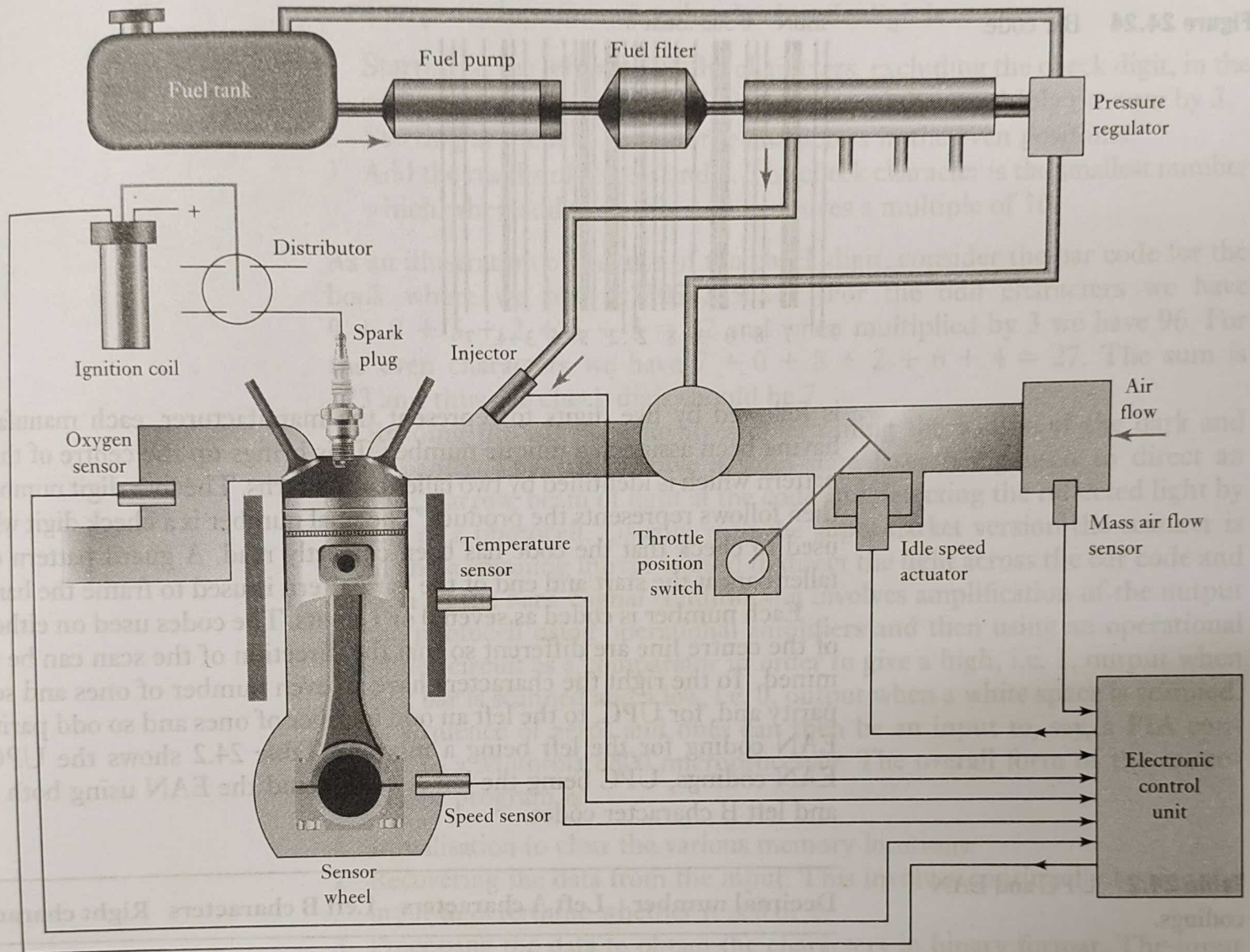


Figure 24.23 An engine management system.

flow. The oxygen sensor is generally a closed-end tube made of zirconium oxide with porous platinum electrodes on the inner and outer surfaces. Above about 300°C the sensor becomes permeable to oxygen ions and so gives a voltage between the electrodes.